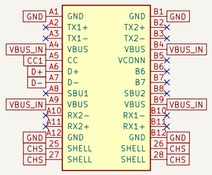
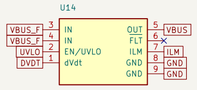
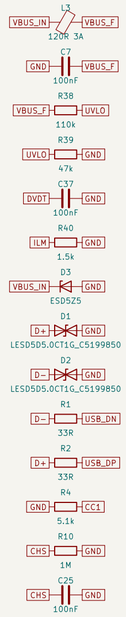
## USB-C plug and input protection

J1 is a plug: only A5 carries CC, and B5 (VCONN) stays open.
U14 takes the ring of a hot plug on its input, which is rated
for 20 V, then raises VBUS at 0.42 V/ms. It passes USB power
above 4.0 V, stops under 3.7 V and limits it to 1.37 A.

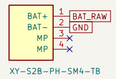


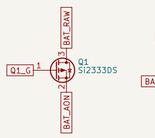



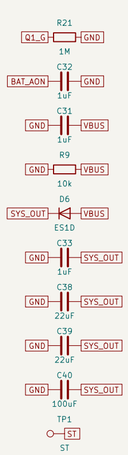



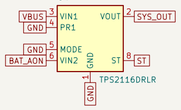

## Battery input and power mux

Q1 blocks a reversed battery, U4 passes the higher of USB and
the battery; its select and mode pins are on ground. C38 to C40,
with C1, feed the converter while it changes over. D6 keeps VBUS
within a diode's drop of the mux's output. R9 empties VBUS.



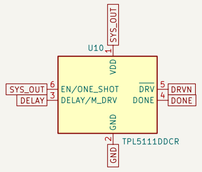



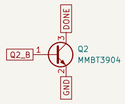


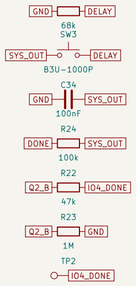


## Wake timer and DONE stage

Power off: drive IO4 high, then low.
DONE rises when Q2 lets go.
TP2 is on IO4, ahead of R22.

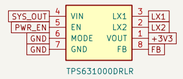


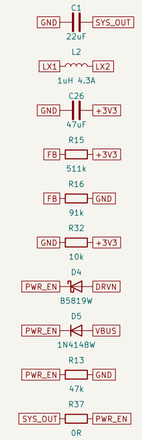




## 3.3 V buck-boost converter

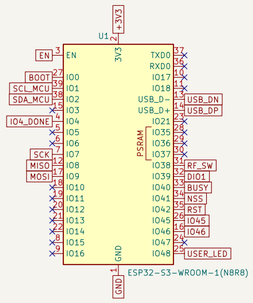
EN is high while the timer drives DRVN
or while USB power is present.
R37 keeps it high on battery, for firmware that
never signals DONE. The timed order leaves it out.
D5 is silicon; a Schottky leaks when warm.



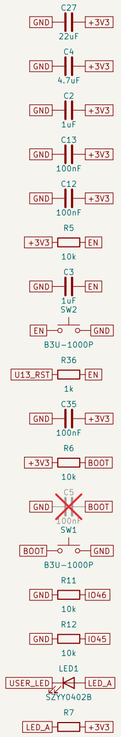

## ESP32-S3 module

IO35 to IO37 belong to the module's
PSRAM and are never used.
U13 holds EN low while +3V3 is under 2.93 V.
R36 keeps its output under 5 mA as C3 discharges.



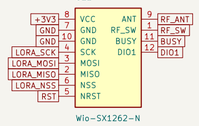


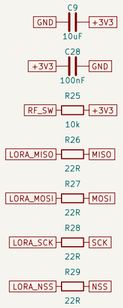





## LoRa module

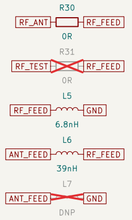
U11 is Seeed's 113991436, the Wio-SX1262 without
the antenna connector. It is sent to the assembler.




## 868 MHz feed and antenna

Production fits R30 only.
The antenna-test order fits R31 and J3 instead.
L5, L6 and L7 are tuned on the VNA.
TL1 is the antenna's printed tuning line.







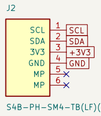


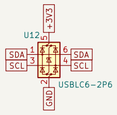


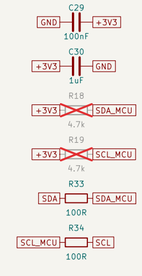


## I2C sensor connector

R18 and R19 are footprints only: the
sensor board carries the pull-ups.
R33 and R34 slow the edges on the cable.




Generated by tools/build.py from tools/design.py: change that and rebuild, do not edit by hand
Rebuilt in KiCad from the PCB_V3 fabrication data with the review changes applied

| | |
|---|---|
| Sheet: / | |
| File: lora-sensor.kicad_sch | |
| **Title: LoRa sensor node** | |
| Size: A2 | Date: 2026-11-10 |
| KiCad E.D.A. 10.0.8 | **Rev: 3.1** |
| | Id: 1/1 |